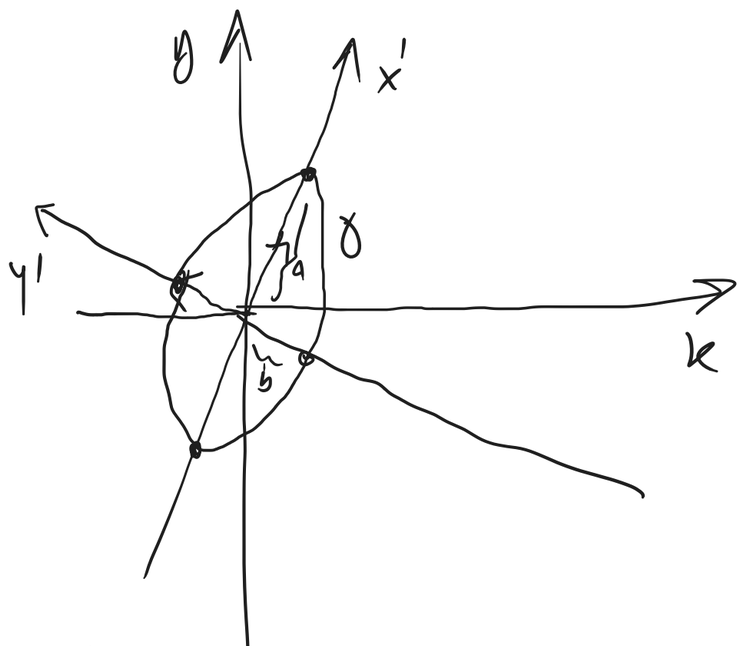


asse x': $y = 2x$

γ ellisse con semiassi

$a = 2 \quad b = 1$

Determinare eq. di γ nel sistema Oxy

$\gamma: \frac{X'^2}{a^2} + \frac{Y'^2}{b^2} = 1$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = {}^t P \begin{bmatrix} x \\ y \end{bmatrix}$$

$$P = [\vec{v}_1, \vec{v}_2] = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\vec{v}_1 = \text{vers}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

ass x': $y = 2x$

$$\begin{cases} x' = \frac{1}{\sqrt{5}}(x + 2y) \\ y' = \frac{1}{\sqrt{5}}(-2x + y) \end{cases}$$

$\gamma: \frac{x'^2}{4} + y'^2 = 1$

sostituisco e ottengo γ di γ rispetto a x, y - -

$2a_{12} = 24$

Caso $a_{12} \neq 0$

$\gamma: 9x^2 + 24xy + 16y^2 - 4x + 3y - 5 = 0$ γ NON deg.

$$A = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} \quad P(\lambda) = |A - \lambda I| = (9-\lambda)(16-\lambda) - 144 = \lambda^2 - 25\lambda = \lambda(\lambda - 25) \quad \lambda_1 = 25, \ \lambda_2 = 0$$

$\det A = 0 \Rightarrow$ parabola

$$P = [\vec{v}_1, \vec{v}_2] = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$\vec{v}_1: -16x + 12y = 0$

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix} \quad \begin{cases} x = \frac{1}{5}(3x' - 4y') \\ y = \frac{1}{5}(4x' + 3y') \end{cases}$$

$\gamma = \lambda_1 x'^2 + \lambda_2 y'^2$

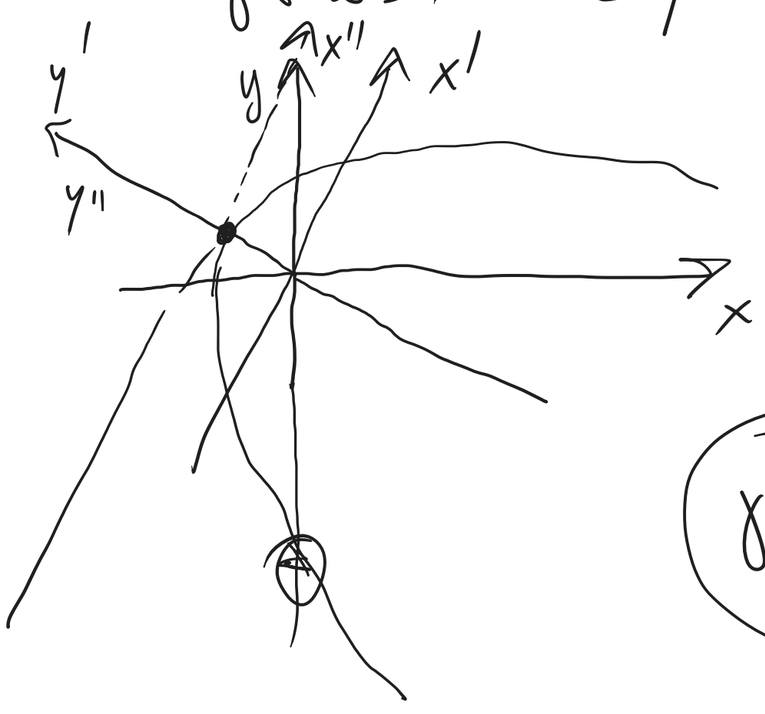
$\gamma: 9x^2 + 24xy + 16y^2 - 4x + 3y - 5 = 0$

$\gamma: 25x'^2 - \frac{4}{5}(3x' - 4y') + \frac{3}{5}(4x' + 3y') - 5 = 0$

$\gamma: 25x'^2 + 5y' - 5 = 0$ $\qquad \gamma: 5x'^2 + y' - 1 = 0$

asse x': $y = \frac{4}{3}x$

$\gamma: y' = 1 - 5x'^2$



$$\begin{cases} x'' = x' \\ y'' = y' - 1 \end{cases}$$

$\gamma: y'' = -5x''^2$